Enrico Castelli

M.Sc. Student in Automation and Control Engineering

Vignola (MO), Italy
enricocastelli03@gmail.com
enrico-castelli-4a3b16323
chettante
Born 7 May 2003, Modena (Italy) • Italian citizen



Profile

Master's student in Automation and Control Engineering at Politecnico di Milano (current weighted average **29/30**), with a strong interest in mobile robotics, perception and control. Hands-on experience in autonomous field robotics (AIRLab FRE team), computer vision, graph SLAM, vision-based drone tracking and embedded neural networks.

Education

2025–present **M.Sc. in Automation and Control Engineering**, *Politecnico di Milano*, Milan, Italy, *Laurea Magistrale*, class LM-25
Enrolled 20/08/2025, 2nd year. Current weighted average: **29/30** (106.33/110), 60 ECTS passed. Planned thesis: 20 ECTS.

2022–2025 **B.Sc. in Automation Engineering**, *Politecnico di Milano*, Milan, Italy, Final grade: **100/110**

2017–2022 **Scientific High School Diploma**, *Liceo Scientifico "M. Allegretti"*, Vignola (MO), Italy, Final grade: **100/100**

M.Sc. Transcript of Records

Course	ECTS	Grade	Date
Dynamics of Mechanical Systems	10	30/30	01/2026
Model Identification and Data Analysis	10	27/30	01/2026
Computer Aided Manufacturing	10	30/30	02/2026
Advanced and Multivariable Control	10	27/30	06/2026
Automation and Control in Electric and Hybrid Vehicles	5	30 cum laude	06/2026
Perception, Localization and Mapping for Mobile Robots	5	30 cum laude	06/2026
Dynamics of Electrical Machines and Drives	10	30/30	07/2026
Weighted average: 29/30 (106.33/110) – 60 ECTS, as of 10/10/2026			

In progress Artificial Neural Networks and Deep Learning; Automation and Control in Autonomous Vehicles; Automation and Control Laboratory; Collaborative, Intelligent and Humanoid Robotics; Image Analysis and Computer Vision; Robotics Laboratory; Software Engineering (for Automation); Advanced Topics in Automation and Control Engineering; Thesis (20 ECTS).

Research & Project Experience

08/2026–10/2026 **Drone Chase: vision-based tracking of a flying target**, *Aerial robotics team project*
Simulated follower quadrotor that finds, tracks and re-acquires a leader drone using only an on-board camera model (Python, telekyb3/GenoM3, Gazebo). Developed the camera field-of-view model, the camera-based velocity estimation, the P + feedforward tracking controller and the search strategy within a four-state mission state machine (project page, code).

03/2026–06/2026 **Member, AIRLab FRE Team (Field Robot Event)**, *AIRLab, Politecnico di Milano*, Milan, Italy

International autonomous agricultural robotics competition; took part in the 23rd Field Robot Event in Bernburg, Germany (16–18 June 2026). Developed marker recognition and localization using OpenCV histogram back-projection and graph SLAM within a ROS 2 system (project page and technical report).

10/2025–present **Member, AI-ARMS Team**, *Automation Engineering Association, Politecnico di Milano*, Milan, Italy

Built and teleoperated (leader–follower) an open-source, 3D-printed SO-101 robotic arm. Trained an ACT imitation-learning policy with LeRobot on 50 teleoperated demonstrations for autonomous pick-and-place without teleoperation (project page).

02/2025–06/2025 **Student Competition: range maximization of electric vehicles**, *Politecnico di Milano*, Milan, Italy

Extracurricular course: designed a control strategy to improve the energy efficiency of an RC car using MATLAB and Simulink; teamwork, communication and leadership in a project team.

Additional Training

03/2026 **Elements of Neural Networks for Microcontrollers and Edge AI**, *Politecnico di Milano, taught by STMicroelectronics*

Trained image-recognition networks in Keras (Google Colab) and deployed them on an STM32 microcontroller with ST software tools.

Certifications

07/2026 **Learn Object-Oriented Programming in C++**, *Educative*

Online course, 6 hours. Certification ID: G9UXRNM104

07/2026 **Learn C++: The Complete Course for Beginners**, *Educative*

Online course, 10 hours. Certification ID: 0812FINJCK

06/2026 **Certificate of Participation – 23rd Field Robot Event**, *Hochschule Osnabrück & TU Braunschweig*, Bernburg (Saxony-Anhalt), Germany

International field robotics competition, 16–18 June 2026, as a member of the Politecnico di Milano (AIRLab) team.

07/2025 **TOEIC Listening & Reading**, *ETS Global*

Score **980/990** (Listening 495, Reading 485), CEFR level C1. Valid until 07/2027.

06/2025 **Open Badge – Student competition: control for minimizing the energy consumption of an electric vehicle**, *Politecnico di Milano*

Digital badge issued by Politecnico di Milano.

Skills

Programming	MATLAB, Simulink, Python, C++ (incl. object-oriented programming), LaTeX
Robotics & Vision	ROS 2, GenoM3 / telekyb3, Gazebo, OpenCV, graph SLAM, localization and mapping
Machine Learning	Imitation learning (LeRobot, ACT), Keras / TensorFlow, neural networks on microcontrollers (STM32)
Control	Multivariable control, system identification, electric drives control

Languages

Italian	Mother tongue	
English	C1	*TOEIC Listening & Reading: 980/990*

Interests

Sport	Competitive basketball player since age 8 — teamwork, communication and a positive team environment
Other	Robotics, travel, trekking, skiing